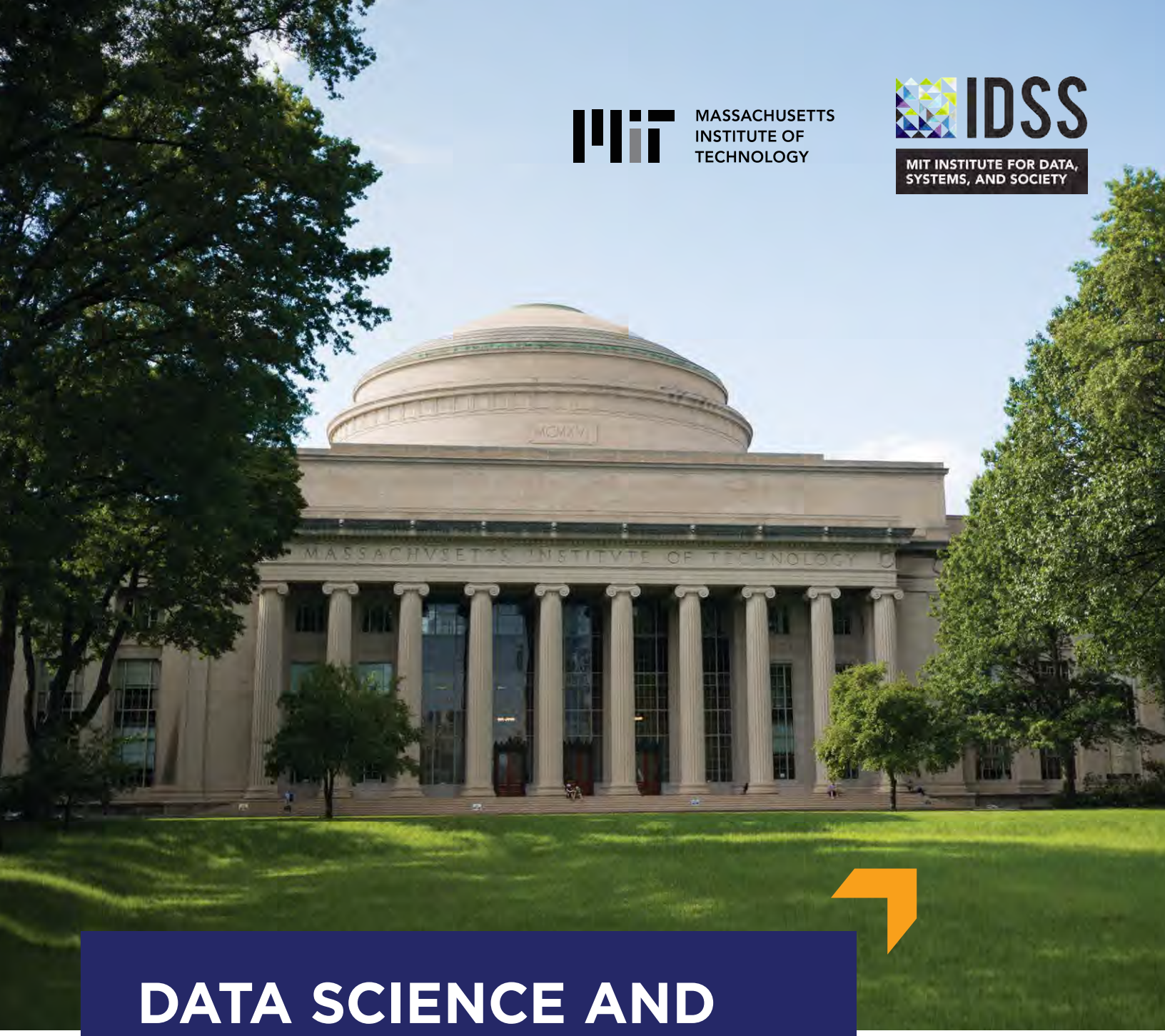


MASSACHUSETTS INSTITUTE OF TECHNOLOGY

IDSS
MIT INSTITUTE FOR DATA, SYSTEMS, AND SOCIETY

DATA SCIENCE AND MACHINE LEARNING: MAKING DATA-DRIVEN DECISIONS

Become a data-driven decision maker with the 12-week online program delivered by MIT faculty

Now Featuring ChatGPT and Generative AI Modules

ABOUT

MIT IDSS

Education and research at MIT Institute for Data, Systems, and Society (IDSS) are undertaken with the aim to provide solutions to complex societal challenges by understanding and analyzing data. The institute is thus committed to the development of analytical methods that can be applied to diverse areas such as finance, energy systems, urbanization, social networks, and health.

MIT IDSS embraces the collision and synthesis of ideas and methods from analytical disciplines including statistics, Data Science, information theory and inference, systems and control theory, optimization, economics, human and social behavior, and network science. These disciplines are relevant both for understanding complex systems and for presenting design principles and architectures that allow for the systems' quantification and management.

MISSION

The mission of MIT IDSS is to advance education and research by using state-of-the-art analytical methods in information and decision systems, statistics, Data Science and social sciences. These methods are leveraged to address complex societal challenges in a diverse set of areas such as finance, energy systems, urbanization, social networks, and health.

Technological advances in areas such as smart sensors, big data, communications, computing, and social networking are rapidly scaling the size and complexity ofi nterconnected systems and networks, and generating data that can lead to new insights and understanding. Research at IDSS aims tounderstand and analyze data from across these systems, which present unique andactionable insights.

ABOUT

THE PROGRAM

Demand for professionals skilled in Data, Analytics, and Machine Learning is exploding. According to a report by the U.S. Bureau of Labor Statistics, the demand for Data Science is set to increase, creating 11.5 million new data-driven jobs by 2026. Data scientists bring value to organizations across industries because they are able to solve complex challenges with data and drive important decision-making processes.

MIT IDSS understands the power of uncovering the true value of your data and has created a variety of online courses and programs to take your data analytics skills to the next level. Whether you are looking to break into the field, seeking career development opportunities, or simply want to provide more valuable insights, these offerings will teach you to harness data in new and innovative ways.



PROGRAM BENEFITS

- Learn online from 11 award-winning MIT faculty and instructors
- Fuel your career transition with CV & LinkedIn review sessions, and demonstrate your Data Science expertise by building a portfolio of 3 real-life projects and 50+ case studies
- Work in a robust collaborative environment and network with peers in Data Science and Machine Learning
- Get a Certificate of Completion from MIT IDSS
- Get mentorship from industry experts on the applications of concepts taught by the faculty
- Earn 8 Continuing Education Units (CEUs) upon program completion

PROGRAM STRUCTURE

30+ hours of recorded lectures from MIT faculty, mentored sessions with experts and hands-on projects, structured as follows:

WEEK 1 & 2	**Foundations of Data Science**
WEEK 3	**Learning Break**
WEEK 4	**Making Sense of Unstructured Data**
WEEK 5	**Learning Break with Hands-On Masterclass 1**
WEEK 6	**Regression and Prediction**
WEEK 7	**Learning Break with Hands-On Masterclass 2**
WEEK 8	**Classification and Hypothesis Testing**
WEEK 9	**Learning Break with Hands-On Masterclass 3**
WEEK 10	**Deep Learning**
WEEK 11	**Recommendation Systems**
WEEK 12	**Networking and Graphical Models**
SELF-PACED	**2 Modules on ChatGPT and Generative AI**

WHO IS THIS
PROGRAM FOR?

- Data Scientists, Data Analysts, and working professionals who wish to extract actionable insights from large volumes of data
- Due to the broad nature of the program, it is suited for both early-career professionals and senior managers, including technical managers, business intelligence analysts, data science managers, data science enthusiasts, IT practitioners, management consultants, and business managers
- Those with some academic/professional training in applied mathematics and statistics will find it easier to learn. However, all participants can make the most out of this program by learning these topics through Great Learning's support

UPON SUCCESSFUL COMPLETION OF THE PROGRAM
YOU WILL BE ABLE TO

- Understand the intricacies of Data Science techniques and their applications to real-world problems
- Implement various Machine Learning techniques to solve complex problems and make data-driven business decisions
- Explore two major realms of Machine Learning - Deep Learning, and Neural Networks, and how they can be applied to areas such as Computer Vision
- Choose how to represent your data when making predictions
- Understand the theory behind recommendation systems and explore their applications to multiple industries and business contexts
- Build an industry-ready portfolio of projects and demonstrate your ability to extract business insights from data

PROGRAM

CURRICULUM

The program is 12 weeks long:

WEEK 1-2

Foundational Courses: Python and Statistics for Data Science

Python for Data Science

- NumPy
- Pandas
- Data Visualization

Case Study 1: *FIFA World Cup Analysis*
Assessment: *MovieLens Project*

Stats for Data Science

- Descriptive Statistics
- Inferential Statistics

Case Study 2: *Fitness Product Customer Footfall Analysis*
Assessment: *MovieLens Project*

WEEK 3

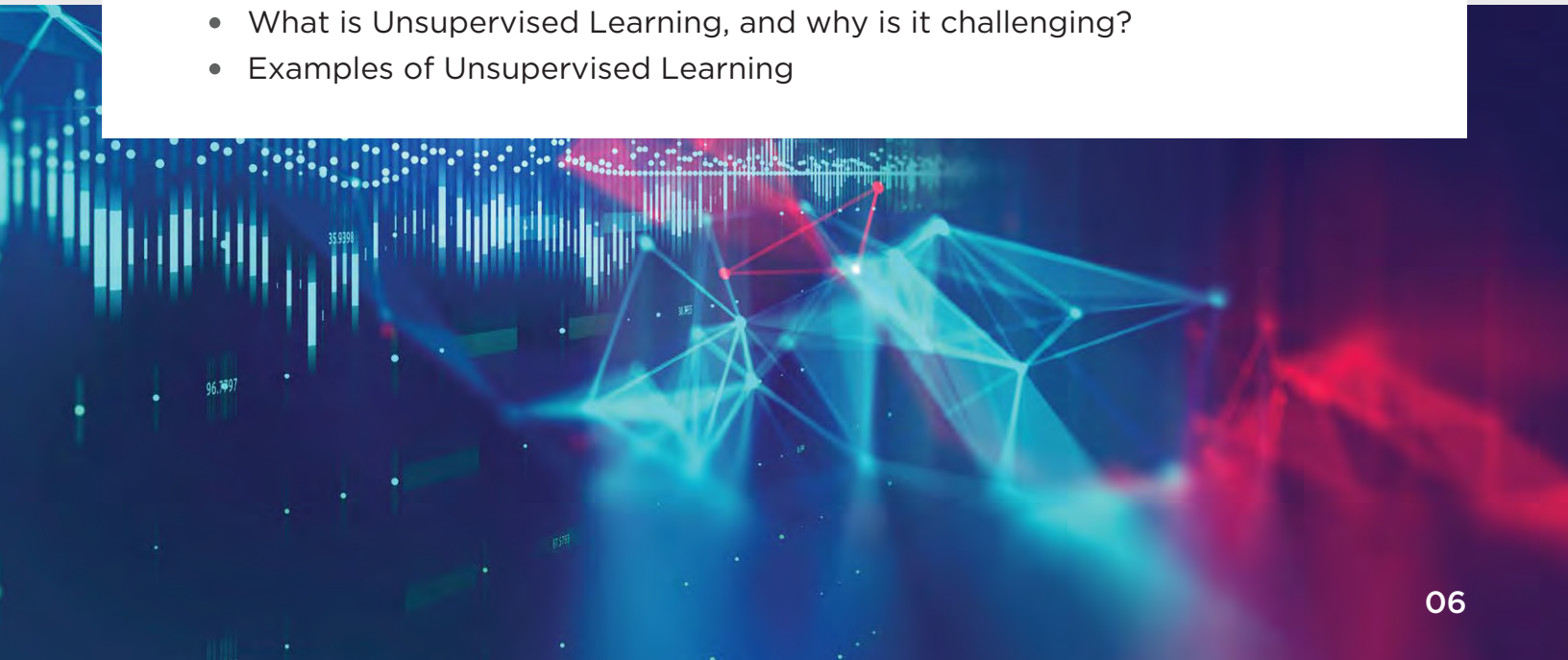
LEARNING BREAK

WEEK 4

Making Sense of Unstructured Data

Introduction

- What is Unsupervised Learning, and why is it challenging?
- Examples of Unsupervised Learning



Clustering

- What is Clustering?
- When to use Clustering?
- K-means Preliminaries
- The K-means algorithm
- How to Evaluate Clustering?
- Beyond K-means: What really makes a Cluster?
- Beyond K-means: Other notions of Distance
- Beyond K-means: Data and Pre-Processing
- Beyond K-means: Big Data and Non-parametric Bayes
- Beyond Clustering

Case Study 1: *Genetic Codes*
Case Study 2: *Finding Themes in the Project Description*

Spectral Clustering, Components, and Embeddings

- What if we do not have features to describe the data or not all are meaningful?
- Finding the principal components in data and applications
- The magic of Eigenvectors I
- Clustering in graphs and networks
- Features from graphs: The magic of Eigenvectors II
- Spectral Clustering
- Modularity Clustering
- Embeddings: New features and their meaning

Case Study 3: *PCA: Identifying Faces*
Case Study 4: *Spectral Clustering: Grouping News Stories*

WEEK 5

LEARNING BREAK
With Hands-On Masterclass 1

WEEK 6

Regression and Prediction

Classical Linear and Non-Linear Regression and Extensions

- Linear Regression with one and several Variables
- Linear Regression for Prediction
- Linear Regression for Causal Inference
- Logistic and other types of Non-Linear Regression

Case Study 1: *Predicting Wages 1*
Case Study 2: *Gender Wage Gap*

Modern Regression with High-Dimensional Data

- Making good predictions with high-dimensional data
- Avoiding Overfitting by Validation and Cross-Validation
- Regularization by Lasso, Ridge, and their Modifications
- Regression Trees, Random Forest, Boosted Trees

Case Study 3: *Do Poor Countries Grow Faster than Rich Countries?*

The Use of Modern Regression for Causal Inference

- Randomized Control Trials
- Observational Studies with Confounding

Case Study 4: *Predicting Wages 2*
Case Study 5: *The Effect of Gun Ownership on Homicide Rates*

WEEK 7

LEARNING BREAK
With Hands-On Masterclass 2

WEEK 8

Classification and Hypothesis Testing

- What are anomalies? What is fraud? Spams?
- Binary Classification: False Positive/Negative, Precision/Recall, F1-Score
- Logistic and Probit Regression: Statistical Binary Classification
- Hypothesis Testing: Ratio Test and Neyman-Pearson p-values: Confidence
- Support Vector Machine: Non-Statistical Classifier
- Perceptron: Simple Classifier with Elegant Interpretation

Case Study 1: *Logistic Regression: The Challenger Disaster*

WEEK 9

LEARNING BREAK
With Hands-On Masterclass 3

WEEK 10

Deep Learning

- What is Image Classification? Introduce ImageNet and show examples
- Classification using a Single Linear Threshold (Perceptron)
- Hierarchical Representations
- Fitting parameters using back-propagation
- Non-Convex Functions
- How interpretable are its features?
- Manipulating Deep Nets (Ostrich Example)
- Transfer Learning
- Other applications I: Speech Recognition
- Other applications II: Natural Language Processing

Case Study 1: *Decision Boundary of a Deep Neural Network*

WEEK 11

Recommendation Systems

Recommendations and Ranking

- What does a Recommendation System do?
- What is the Recommendation Prediction Problem? And what data do we have?
- Using Population Averages
- Using Population Comparisons and Ranking

Case Study 1: *Recommending Movies*

Collaborative Filtering

- Personalization using Collaborative Filtering using similar users
- Personalization using Collaborative Filtering using similar items
- Personalization using Collaborative Filtering using similar users and items

Case Study 2: *Recommend New Songs to the Users Based on Their Listening Habits*

Personalized Recommendations

- Personalization using Comparisons, Rankings, and Users-Items
- Hidden Markov Model / Neural Nets, Bipartite Graph, and Graphical Model
- Using Side-Information
- 20 questions and Active Learning
- Building a system: Algorithmic and System Challenges

Case Study 3: *Make New Product Recommendations*

WEEK 12

Networking and Graphical Models

Introduction

- Introduction to Networks
- Examples of Networks
- Representation of Networks

Case Study 1: *Navigation / GPS*
1.1: *Kalman Filtering: Tracking the 2D Position of An Object When Moving with Constant Velocity*
1.2: *Kalman Filtering: Tracking the 3D position of An Object Falling Due to Gravity.*

Networks

- Centrality Measures: Degree, Eigenvector, and Page-Rank
- Closeness and betweenness Centrality
- Degree Distribution, Clustering, and Small World
- Network models: Erdos-Renyi, Configuration model, Preferential Attachment
- Stochastic models on networks for the spread of viruses or ideas
- Influence Maximization

Case Study 2: *Identifying New Genes that Cause Autism*

Graphical Models

- Undirected Graphical models
- Ising and Gaussian models
- Learning Graphical models from Data
- Directed Graphical models
- V-structures, “explaining away,” and learning Directed Graphical models
- Inference in Graphical models: Marginals and Message Passing
- Hidden Markov Model (HMM)
- Kalman Filter

SELF-PACED MODULES WITH AN OPTIONAL MASTERCLASS

Demystifying ChatGPT and its Applications

- Overview of ChatGPT and OpenAI
- Implications for Work, Business and Education
- Prompt Engineering for Fine-Tuning Outputs

ChatGPT: The Development Stack

- Mathematical Fundamentals for Generative AI
- Transformer Models: Generative AI for Natural Language
- Hands-On ChatGPT Prototype Creation

FACULTY



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Program Faculty Director, IDSS, MIT



John N. Tsitsiklis
Clarence J. Lebel Professor, EECS, MIT



Devavrat Shah
Professor, EECS and IDSS, MIT



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Jonathan Kelner
Professor, Mathematics, MIT



Kalyan Veeramachaneni
Principal Research Scientist at the Laboratory for Information and Decision Systems, MIT



Guy Bresler
Associate Professor, EECS and IDSS, MIT

PROGRAM

MENTORS

The program coaches you to work on industry-relevant projects, with guidance from Data Science and Machine Learning experts via live and personalized learning sessions that give you a practical understanding of core concepts.



Bradford Tuckfield
Founder and Data Science Consultant, Kmbara



Mayan Murray
Senior Data Scientist and UX Consultant, IBM



Amit Agarwal
Senior Data Scientist, Oracle



Xiaojun Su
Data Science Product Manager, Unilever



Andrew Marlatt
Data Scientist, Revenue Expansion, Shopify



Srikanth Pyaraka
Data Science Product Manager, Verizon



Shirish Gupta
Lead Data Scientist, Novartis



Thinesh Pathmanathan
Data Scientist, TD



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Analytics Leader, Global Advanced Analytics, AstraZeneca



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Data Scientist, Amazon



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Senior Data Scientist, Walmart Connect



Juan Castillo
Machine Learning Engineer, SEPHORA



Rohit Dixit
Senior Data Scientist, Siemens Healthineers



Angel Das
Data Science Consultant, IQVIA Asia Pacific



Vanessa Afolabi
Senior Data Scientist, Loblaw Companies Limited



Grivine Ochieng
Lead Data Engineer, Xetova

PROGRAM MANAGER:

YOUR PERSONAL GUIDE

Your Program Manager will assist you through the learning journey to ensure you achieve your learning objectives. They will act as your sole point of contact during the program, supporting you by ensuring you receive the appropriate and timely assistance from the ecosystem. Along with monitoring your progress, they will be there to give you the necessary encouragement to ensure your success.

CERTIFICATE



The image is for illustrative purposes only. The actual certificate may be subject to change at the discretion of the university.

APPLICATION PROCESS

Step-1

Application Form

Register by completing the online application form.

Step-2

Application Screening

Your application will be reviewed to determine whether you're eligible for this program.

Step-3

Join the Program

If selected, you will receive an offer for the upcoming cohort. Secure your seat by paying the fee.

APPLICATION & FEE DETAILS

Program Duration: **12 weeks**

Fees: **USD 2,300**

The MIT IDSS Data Science and Machine Learning Program, with a curriculum developed and taught by MIT faculty, is delivered in collaboration with

Great Learning
POWER AHEAD

Great Learning is a leading global ed-tech company for professional and higher education. It offers comprehensive, industry-relevant programs across various cutting-edge Technology, Data, and Business domains. These programs are developed in collaboration with the world's foremost academic institutions such as Stanford Executive Education, MIT Professional Education, Wharton Online, The University of Texas at Austin, Northwestern School of Professional Studies, National University of Singapore, Deakin University, IIT Madras, IIT Roorkee, IIIT-Delhi, Great Lakes Institute of Management, and more. They are constantly reimagined and revamped to address the dynamic needs of the industry. Offered in blended, classroom and purely online modes, these programs are delivered with the help of expert mentors and highly qualified faculty. Great Learning is on a mission to enable transformative learning and career success in the digital economy and has impacted 9 million+ learners from over 170 countries.



READY TO BECOME A DATA-DRIVEN DECISION MAKER?

APPLY NOW

SPEAK TO A PROGRAM ADVISOR

+1 617 539 7216

HAVE QUESTIONS ABOUT THE PROGRAM OR HOW IT FITS IN WITH YOUR CAREER GOALS?

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VISIT OUR WEBSITE

https://www.mygreatlearning.com/mit-data-science-machine-learning-program/